

LUMBAR PUNCTURE — Procedure One-Pager

1 · INDICATIONS

Dx — urgent	Suspected meningitis/encephalitis; SAH w/ negative CT
Dx — non-urgent	CNS malignancy (cytology); NPH; autoimmune (MS, GBS, CNS vasculitis, paraneoplastic)
Therapeutic	IIH/pseudotumor (also dx); intrathecal med or contrast

2 · CONTRAINDICATIONS

Absolute	Overlying skin infection / suspected epidural abscess at site
Relative	Space-occupying lesion / Arnold-Chiari (CT ± NSGY first) · plt <50K · INR ≥1.5 · therapeutic anticoag · prior lumbar surgery

CT before LP only if: AMS, focal deficit, papilledema, seizure <1 wk, or immunocompromise (Hasbun NEJM 2001). **Never delay antibiotics** for CT/LP in suspected bacterial meningitis — cultures + empiric abx first.

3 · POSITION & LANDMARKS

Sitting upright	Lateral decubitus
Lean over table, feet on stool, knees > hips. Easier landmarks. No OP possible.	Handed-side down; stack ankles/ knees/shoulders; chin tucked, knees up. Required for OP.

- **Palpation:** iliac crest → ~L4; mark midline at L3–L4, L4–L5, or L2–L3.
- **Stay L2–L5** (below conus medullaris).
- **US (recommended):** low-freq probe from sacrum up; transverse = midline, longitudinal = interspaces. Improves success, fewer passes (SHM 2019).

4 · TECHNIQUE

- **Time-out** w/ RN (name/DOB/procedure/site/allergies) before opening sterile supplies.
- **Prep:** iodine-povidone ×3, let dry (preferred over CHG). Fenestrated drape landscape.
- **Anesthesia:** 1% lidocaine w/o epi; dermal wheal ≤0.5 cc (don't obscure marks), then tract.
- **Needle:** midline, toward umbilicus, **stylet in**; advance slowly, remove stylet frequently for CSF. Bone → withdraw to subQ, redirect midline.

SAFETY STOP: never inject through the spinal needle; never advance without the stylet. Can't stay midline → re-mark (US) & reposition, don't fan blindly.

Use an atraumatic (pencil-point) needle — strong rec for ALL patients; halves PDPH (~11% → ~4%), equal success (BMJ/Lancet 2018). 22 g balances PDPH/flow/OP. Replace stylet before withdrawal. **Bed rest & IV fluids do NOT prevent PDPH.**

5 · OPENING PRESSURE & COLLECTION

- **OP (lateral decubitus only):** normal **10–25 cm H₂O**. Elevated → straighten legs (drops ~5 cm H₂O). Upright falsely elevates.
- **Volume:** routine (chem, cell count, viral, bacterial cx) ~8–12 mL over 4 tubes; **cytology + fungal/AFB ≥30 mL**.
- **Slow flow:** rotate needle in 30° increments.
- **Closure:** replace stylet → withdraw needle (stylet in) → hemostasis → dressing.

CSF STUDIES BY INDICATION

Suspect	Send
Meningitis/enceph	Cell count+diff, protein, glucose, Gram stain+cx, HSV 1/2 PCR, OP
IIH	OP (>25 = classic) + routine
Malignancy	Cytology (≥30 mL), flow if heme
MS/autoimmune	Oligoclonal bands, IgG index
Immunocomp.	+ Fungal/AFB cx (≥30 mL), crypto Ag

6 · MASTERY-LEARNING THRESHOLDS

Component	MPS (sim)	UPS (clinical)
Mistakes	<12	<3
Global skills	≥beginner (>2)	≥proficient
Entrustment	full supervision	indirect supervision

PDPH prevention recap: atraumatic needle + smaller gauge + replace stylet. Risk factors for PDPH: younger age, female, low BMI, prior PDPH, cutting needle.

Adapted from TeachIM.org Lumbar Puncture (Fainstad et al., Dec 2023; sec. ed. Cool), CC BY-NC 4.0 — non-commercial educational use. Optimized June 2026; currency verified (BMJ/Lancet 2018 atraumatic-needle rec, SHM 2019 US, ABN 2018 antithrombotic, Hasbun 2001 all current). Clinician-review before teaching. Not medical advice.